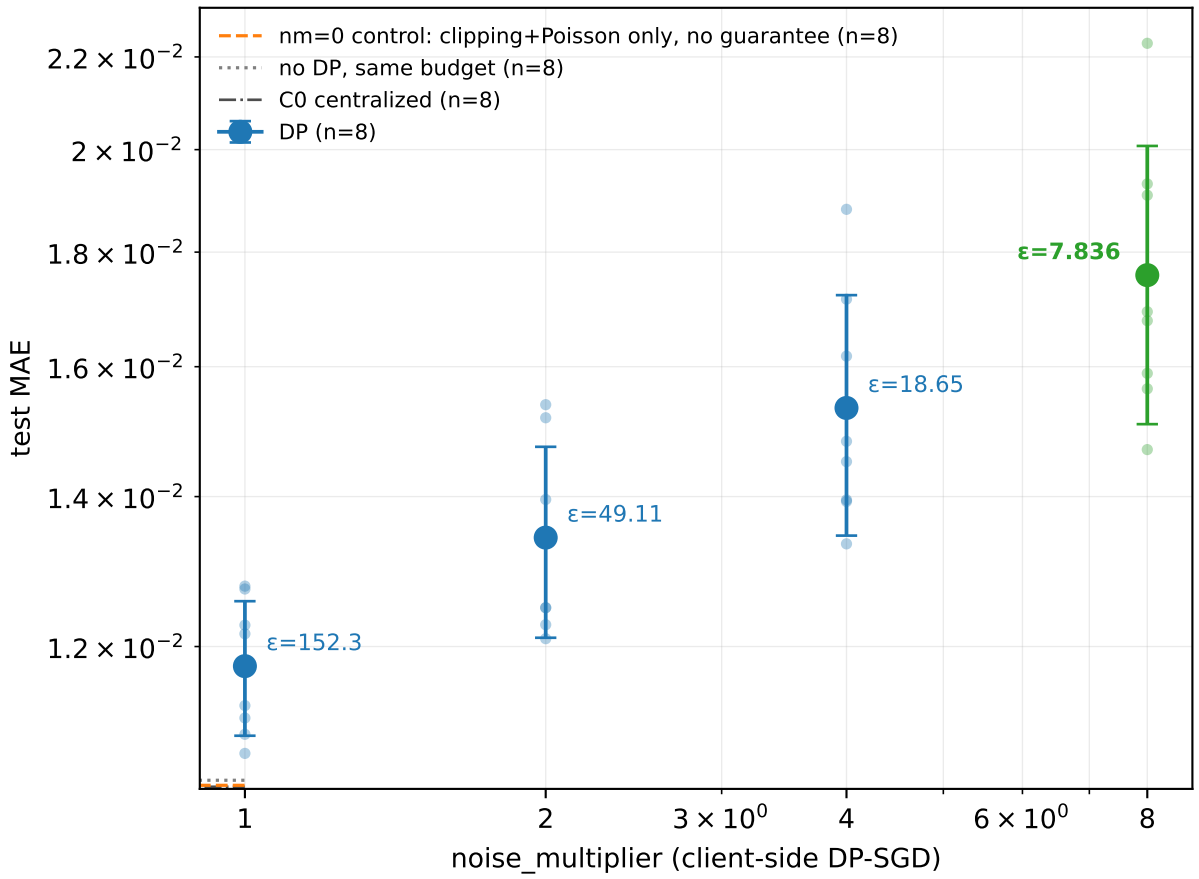
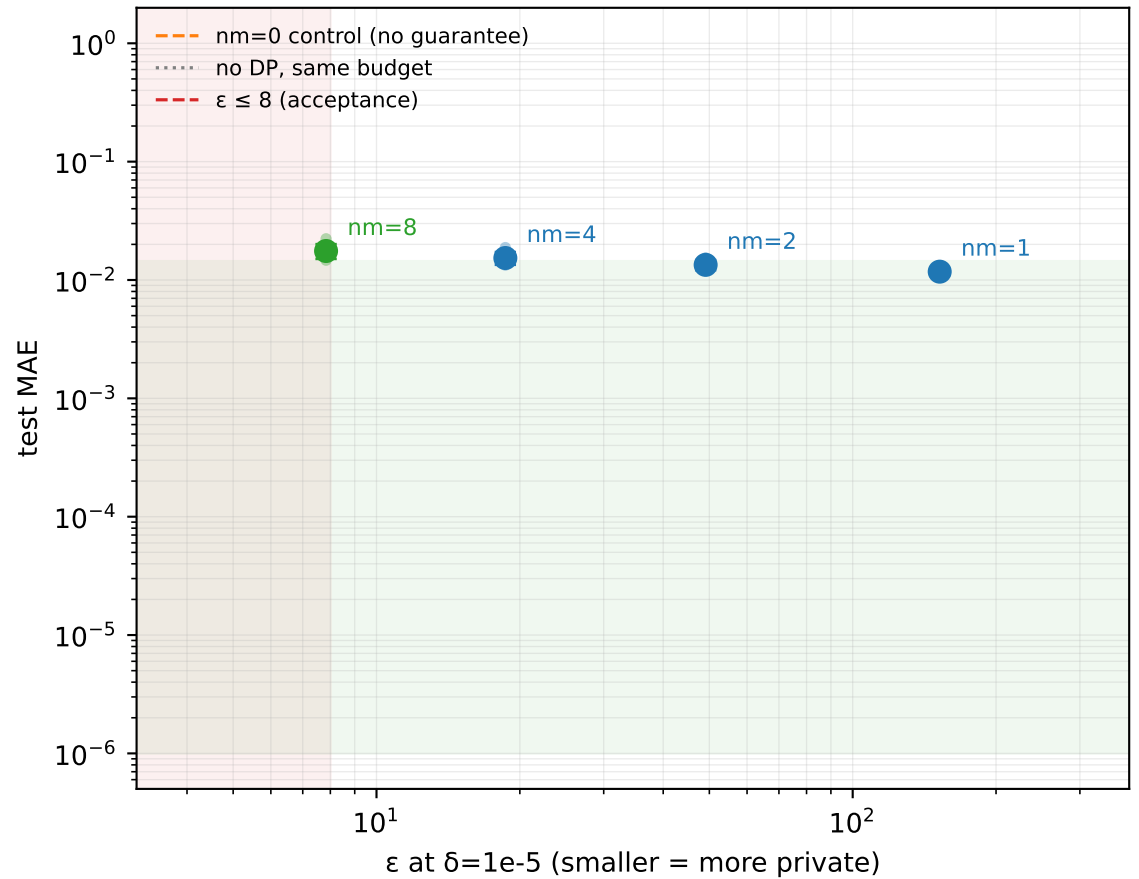

M4: utility vs noise (le=3); ε annotated per point
green = reaches the ε budget
nm=0 control: clipping+Poisson only, no guarantee (n=8)
no DP, same budget (n=8)
C0 centralized (n=8)
DP (n=8)
ε=152.3
ε=49.11
ε=18.65
ε=7.836
test MAE
noise_multiplier (client-side DP-SGD)


Privacy-utility trade-off; the shaded region is what the
acceptance criterion actually allows
nm=0 control (no guarantee)
no DP, same budget
ε ≤ 8 (acceptance)
nm=8
nm=4
nm=2
nm=1
test MAE
ε at δ=1e-5 (smaller = more private)